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DIFERENCIABILIDAD

* Ejercicio 18 = Hallar matriz jacobiana de los campos, y los gradientes

(a) $\vec{F}(x, y) = (3x^2y, x-y) \rightarrow$ Diferenciable ya que f_1 y f_2 diferenciables

• $f_1(x, y) = 3x^2y$

$$\frac{\partial f_1}{\partial x}(x, y) = 6xy \quad \frac{\partial f_1}{\partial y}(x, y) = 3x^2 \quad \nabla f_1(x, y) = (6xy, 3x^2)$$

• $f_2(x, y) = x - y$

$$\frac{\partial f_2}{\partial x}(x, y) = 1 \quad \frac{\partial f_2}{\partial y}(x, y) = -1 \quad \nabla f_2(x, y) = (1, -1)$$

$$J_f = \begin{pmatrix} 6xy & 3x^2 \\ 1 & -1 \end{pmatrix}$$

(b) $G(x) = (x^2 + 1, 2x)$, $g_1 = x^2 + 1$, $g_2 = 2x$

$$\frac{\partial g_1}{\partial x}(x) = 2x \quad \frac{\partial g_2}{\partial x}(x) = 2 \quad J_G = \begin{bmatrix} 2x \\ 2 \end{bmatrix}$$

(c) $h(x, y, z) = xy + z^2x$

$$\frac{\partial h}{\partial x}(x, y, z) = y + z^2 \quad \frac{\partial h}{\partial y}(x, y, z) = x \quad \frac{\partial h}{\partial z}(x, y, z) = 2zx \quad J_h(x, y, z) = \nabla h(x, y, z) = (y + z^2, x, 2zx)$$

(d) $\vec{L}(x, y) = (x^2y, y, x - xy)$ $L_1(x, y) = x^2y$ $L_2(x, y) = y$ $L_3(x, y) = x - xy$

• $L_1(x, y) = x^2y$

$$\frac{\partial L_1}{\partial x}(x, y) = 2xy \quad \frac{\partial L_1}{\partial y}(x, y) = x^2 \quad \nabla L_1(x, y) = (2xy, x^2)$$

• $L_2(x, y) = y$

$$\frac{\partial L_2}{\partial x}(x, y) = 0 \quad \frac{\partial L_2}{\partial y}(x, y) = 1 \quad \nabla L_2(x, y) = (0, 1)$$

$$J_L(x, y) = \begin{pmatrix} 2xy & x^2 \\ 0 & 1 \\ 1-y & x \end{pmatrix}$$

• $L_3(x, y) = x - xy$

$$\frac{\partial L_3}{\partial x}(x, y) = 1 - y \quad \frac{\partial L_3}{\partial y}(x, y) = -x \quad \nabla L_3(x, y) = (1 - y, -x)$$

(f) $f(x, y) = x \sin(2x - y)$

$$\frac{\partial f}{\partial x}(x, y) = \sin(2x - y) + 2\cos(2x - y)x$$

$$\nabla f(x, y) = (\sin(2x - y) + 2\cos(2x - y)x, -x\cos(2x - y))$$

$$\frac{\partial f}{\partial y}(x, y) = x\cos(2x - y)(-1)$$

③ $\vec{N}(x, y, z) = (2xy, x^2 - ze^y)$ $N_1(x, y, z) = 2xy$ $N_2(x, y, z) = x^2 - ze^y$

◦ $N_1(x, y, z) = 2xy$

$\frac{\partial N_1}{\partial x}(x, y, z) = 2y$ $\frac{\partial N_1}{\partial y}(x, y, z) = 2x$ $\frac{\partial N_1}{\partial z}(x, y, z) = 0$ $\nabla N_1(x, y, z) = (2y, 2x, 0)$

◦ $N_2(x, y, z) = x^2 - ze^y$

$\frac{\partial N_2}{\partial x}(x, y, z) = 2x$ $\frac{\partial N_2}{\partial y}(x, y, z) = -ze^y$ $\frac{\partial N_2}{\partial z}(x, y, z) = -e^y$ $\nabla N_2(x, y, z) = (2x, -ze^y, -e^y)$

$J_N(x, y, z) = \begin{pmatrix} 2y & 2x & 0 \\ 2x & -ze^y & -e^y \end{pmatrix}$

* Ejercicio 19 $f(x, y) = x\sqrt{xy}$ y $A = (1, 4) \in D^\circ$

④ Hallar razones suficientes para asegurar diferenciableidad de f en A ↗ ¿Calcular x definición?

$f(1, 4) = 1 \cdot \sqrt{4} = 2$

$\frac{\partial f}{\partial x}(x, y) = \sqrt{xy} + \frac{1}{2\sqrt{xy}} \cdot xy \rightarrow \frac{\partial f}{\partial x}(1, 4) = \sqrt{4} + \frac{4}{2\sqrt{4}} = 2 + 1 = 3$

$\frac{\partial f}{\partial y}(x, y) = \frac{x}{2\sqrt{xy}} \cdot x = \frac{x^2}{2\sqrt{xy}} \rightarrow \frac{\partial f}{\partial y}(1, 4) = \frac{1^2}{2\sqrt{4}} = \frac{1}{4}$

Si es diferenciable: $\lim_{(x, y) \rightarrow (1, 4)} \frac{f(x, y) - [f(1, 4) + \frac{\partial f}{\partial x}(1, 4)(x-1) + \frac{\partial f}{\partial y}(1, 4)(y-4)]}{\sqrt{(x-1)^2 + (y-4)^2}} = 0$

$\lim_{(x, y) \rightarrow (1, 4)} \frac{x\sqrt{xy} - [2 + 3(x-1) + \frac{1}{4}(y-4)]}{\sqrt{(x-1)^2 + (y-4)^2}} = \lim_{(x, y) \rightarrow (1, 4)} \frac{x\sqrt{xy} - 2 - 3x + 3 - \frac{1}{4}y + 1}{\sqrt{(x-1)^2 + (y-4)^2}} = \lim_{(x, y) \rightarrow (1, 4)} \frac{x\sqrt{xy} - 3x - \frac{y}{4} + 2}{\sqrt{(x-1)^2 + (y-4)^2}}$ JND

$f(x, y) = \underbrace{f(A) + \frac{\partial f}{\partial x}(A)(x-x_0) + \frac{\partial f}{\partial y}(A)(y-y_0)}_{L(x)} + \underbrace{\mu(H)}_{\rightarrow 0} \|H\|$

$f(x, y) = 2 + 3(x-1) + \frac{1}{4}(y-4)$

$\underbrace{f(1, 4)}_2 = 2 + 3(1-1) + \frac{1}{4}(4-4) \rightarrow \boxed{2=2}$ ↗ la aproximación lineal es igual a $f(A) = L(A)$

⑤ → Hallar derivadas direccionales máximas, mínimas y nulas $\vec{r} = (a, b)$, $a^2 + b^2 = 1$

$\nabla f(1, 4) = (3, \frac{1}{4})$

$\frac{\partial f}{\partial \vec{r}_{\max}}(1, 4) = \nabla f(1, 4) \cdot \vec{r} = \|\nabla f(1, 4)\| \underbrace{\|\vec{r}\| \cos(\theta)}_1, \rightarrow r_{\max} = \frac{\nabla f(1, 4)}{\|\nabla f(1, 4)\|} = \left(\frac{12}{\sqrt{145}}, \frac{1}{\sqrt{145}}\right)$ $\frac{\partial f}{\partial \vec{r}_{\max}}(1, 4) = 17$

$\frac{\partial f}{\partial \vec{r}_{\min}}(1, 4) = \nabla f(1, 4) \cdot \vec{r} = \|\nabla f(1, 4)\| \underbrace{\|\vec{r}\| \cos(\theta)}_{-1} = -17$ $r_{\min} = -\left(\frac{12}{\sqrt{145}}, \frac{1}{\sqrt{145}}\right)$

$\frac{\partial f}{\partial \vec{r}_0}(1, 4) = \nabla f(1, 4) \cdot \vec{r} = 3a + \frac{1}{4}b = 0$ $\parallel \begin{cases} a^2 + (-12a)^2 = 1 \\ 145a^2 = 1 \rightarrow a = \pm \frac{1}{\sqrt{145}} \end{cases}$ $\vec{r}_0 = \left(\frac{1}{\sqrt{145}}, \frac{-12}{\sqrt{145}}\right)$ $\vec{r}_2 = \left(\frac{-1}{\sqrt{145}}, \frac{12}{\sqrt{145}}\right)$

* Ejercicio 20 $f \in C^1$, $f'(A, (0.6, 0.8)) = 2$ y $f'(A, (0.8, 0.6)) = 5$

→ Hallar máxima derivada direccional de f en A y en qué dirección

$$\vec{r}_1 = (0.6, 0.8) \quad \vec{r}_2 = (0.8, 0.6)$$

$$f'(A, \vec{r}_1) = 2 \quad f'(A, \vec{r}_2) = 5$$

transformación lineal

$$\nabla f(A) = (a, b)$$

• Hallar $\nabla f(A)$ sabemos que

$$\begin{aligned} f'(A, \vec{r}_1) &= \nabla f(A) \cdot \vec{r}_1 = \begin{cases} 0.6a + 0.8b = 2 \\ 0.8a + 0.6b = 5 \end{cases} \Rightarrow \left(\begin{array}{cc|c} 0.6 & 0.8 & 2 \\ 0.8 & 0.6 & 5 \end{array} \right) \xrightarrow{F_2 \rightarrow F_2 - \frac{4}{3}F_1} \left(\begin{array}{cc|c} 0.6 & 0.8 & 2 \\ 0 & -\frac{7}{15} & \frac{7}{3} \end{array} \right) \\ f'(A, \vec{r}_2) &= \nabla f(A) \cdot \vec{r}_2 = \end{aligned}$$

$$\begin{cases} 0.6a + 0.8b = 2 \rightarrow a = \frac{2 - 0.8(-5)}{0.6} = 10 \\ \frac{7}{15}b = \frac{7}{3} \rightarrow \boxed{b = 5} \end{cases}$$

$$\nabla f(A) = (10, -5)$$

$$\frac{\partial f}{\partial r_{\max}}(A) = \|\nabla f(A)\| = \sqrt{(-5)^2 + (10)^2} = \sqrt{125} = 5\sqrt{5} \quad r_{\max} = \frac{\nabla f(A)}{\|\nabla f(A)\|} = \frac{(10, -5)}{\sqrt{125}} = \left(\frac{10}{\sqrt{125}}, \frac{-5}{\sqrt{125}} \right) = \left(\frac{2}{\sqrt{5}}, \frac{-1}{\sqrt{5}} \right)$$

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* Ejercicio 21 $f: D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ diferenciable en $P_0 = (x_0, y_0) \in D^\circ$

→ mostrar $\vec{N}_0 = \left(-\frac{\partial f}{\partial x}(P_0), -\frac{\partial f}{\partial y}(P_0), 1 \right)$ es normal a gráfica f en $Q_0 = (x_0, y_0, f(x_0, y_0))$

⇒ $\vec{N}_0$ normal a $Q_0 \rightarrow \vec{N}_0$ es múltiple de la Normal del plano tangencial en Q_0

• Hallar plano tangencial a Q_0 → plano dif en P_0

$$L(x, y) = f(P_0) + \frac{\partial f}{\partial x}(P_0)(x - x_0) + \frac{\partial f}{\partial y}(P_0)(y - y_0) \quad \rightarrow \text{plano tangencial a } p$$

$$z = f(P_0) + \frac{\partial f}{\partial x}(P_0)(x - x_0) + \frac{\partial f}{\partial y}(P_0)(y - y_0)$$

$$\underbrace{\frac{\partial f}{\partial x}(P_0)(x - x_0)}_{Ax} + \underbrace{\frac{\partial f}{\partial y}(P_0)(y - y_0)}_{By} + \underbrace{-z + f(P_0)}_{Cz + D} = 0$$

vector normal al plano tiene la forma $\left(\frac{\partial f}{\partial x}(P_0), \frac{\partial f}{\partial y}(P_0), -1 \right)$

* Cómo probar diferenciabilidad

$$f(x+h) = f(A) + \frac{\partial f}{\partial x}(A)h + \frac{\partial f}{\partial y}(A)k + \underbrace{\|H\| \mu(H)}_{\xrightarrow{H \rightarrow 0} 0} \text{ donde } \lim_{H \rightarrow 0} \frac{\mu(H)}{\|H\|} = 0 \quad \xrightarrow{\epsilon \in \mathbb{R}} \text{ por def}$$

• Probar que es $C^1 \rightarrow$ entonces es diferenciable

• Probar que No es diferenciable → no es continuo → $\nexists$ derivada por cada $\rightarrow \nexists$ derivadas parciales continuas

* Ejercicio 22 = - Justificar diferenciable en P_0

- Representar gráfico de f , plano tg a sup Q_0 y vector $\vec{N}_0$ aplicado a Q_0

a) $f(x,y) = 5 + 2x - 3y$, $P_0 = (0,0)$

• Probar diferenciable $\rightarrow$ probar clase C^1

$Df = \mathbb{R}^2 \rightarrow$ continua en todo su dom

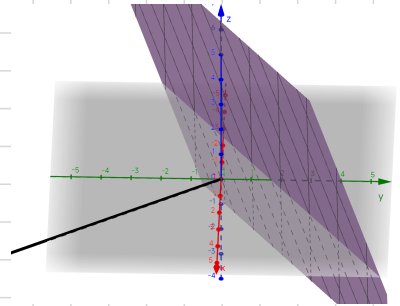
$\frac{\partial f}{\partial x}(x,y) = 2$ $\frac{\partial f}{\partial y}(x,y) = -3 \Rightarrow$ función f es de clase C^1 por lo tanto es diferenciable en su dominio

$\rightarrow P_0 \in Df$

• Hallar gráfico, plano tg, vector tg en $P_0 = (0,0)$

$\Rightarrow$ plano $z = \underbrace{f(0,0)} + \frac{\partial f}{\partial x}(0,0)x + \frac{\partial f}{\partial y}(0,0)y$
 $z = 5 + 2x - 3y$

$\Rightarrow$ Normal $\vec{N}_0 = (2, -3, -1) \rightarrow$ recta normal $\lambda(2, -3, -1) + (0,0,5)$



b) $f(x,y) = x^2 - 2x + y^2$, $P_0 = (-1,2)$

• Hallar Dominio $D_f = \mathbb{R}^2$

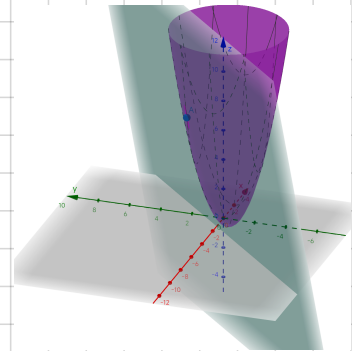
• Verificar clase de función

f es de C^1 por lo que es diferenciable en todo su dom

$\frac{\partial f}{\partial x}(x,y) = 2x - 2$ $\frac{\partial f}{\partial y}(x,y) = 2y \Rightarrow D_{f_x} \subset D_f$ como $P_0 \in D_f$ entonces P_0 es diferenciable
 $\frac{\partial^2 f}{\partial x^2}(-1,2) = -2 - 2 = -4$ $\frac{\partial^2 f}{\partial y^2}(-1,2) = 4$
 $D_{f_y} \subset D_f$

• Hallar plano tangente

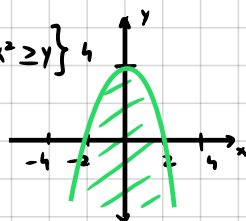
$L(x,y) = \underbrace{f(-1,2)} + \frac{\partial f}{\partial x}(-1,2)(x+1) + \frac{\partial f}{\partial y}(-1,2)(y-2)$
 $z = 7 + (-4)(x+1) + 4(y-2) \Rightarrow \vec{N}_0 = (-4, 4, -1)$



c) $f(x,y) = \sqrt{4-x^2-y^2}$, $P_0 = (1,-1)$

• Ver dominio $D_f = \{(x,y) \in \mathbb{R}^2 / 4-x^2 \geq y^2\}$

$4-x^2-y^2 \geq 0 \rightarrow 4-x^2 \geq y^2$



• Probar diferenciable

$\frac{\partial f}{\partial x}(x,y) = \frac{1}{2\sqrt{4-x^2-y^2}} \cdot -2x$ $\frac{\partial f}{\partial y}(x,y) = \frac{1}{2\sqrt{4-x^2-y^2}} \cdot -2y$
 $D_{f_x} \subset D_f$ $D_{f_y} \subset D_f$

$f(1,-1) = \sqrt{4-1-1} = \sqrt{2}$

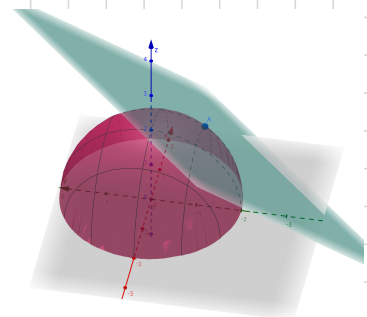
$\frac{\partial f}{\partial x}(1,-1) = -\frac{1}{\sqrt{2}}$

$\frac{\partial f}{\partial y}(1,-1) = \frac{1}{\sqrt{2}}$

• Ecuación plano tg

$L(x,y) = f(1,-1) + \frac{\partial f}{\partial x}(1,-1)(x-1) + \frac{\partial f}{\partial y}(1,-1)(y+1)$

$z = \sqrt{2} - \frac{\sqrt{2}}{2}(x-1) + \frac{\sqrt{2}}{2}(y+1)$



Como f es de clase C^1 es diferenciable

en su dominio $D_f = \{(x,y) \in \mathbb{R}^2 / 4-x^2 \geq y^2\}$

$\rightarrow$ como $P_0 = (1,-1) \in D_f$
 entonces P_0 diferenciable

* Ejercicio 23 = Sabiendo $\checkmark$ $f(x,y)$ tiene plano tg $\pi_1 = 3x + y + 2z = 6$ en $A = (2, 1, z_0)$

↪ Ver si la recta normal al gráfico $f(x,y)$ en A interseca sup $y = x^2$

⇒ Gráfico de f $G_f = (x, y, f(x,y))$

⇒ Plano tg a G_f en $A = G_{tg} = (x, y, L(x,y))$

⇒ Si plano es tg a G_f entonces coinciden en A : $G_f = G_{tg} = A = (2, 1, z_0)$

$$(x, y, f(x,y)) = (x, y, L(x,y)) = (2, 1, z_0)$$

$$\underbrace{(2, 1, f(2,1))}_{G_f} = \underbrace{(2, 1, L(2,1))}_{G_{tg}} = (2, 1, z_0)$$

• Hallar $z = L(x,y)$

• Hallar z_0

$$\pi = 3x + y + 2z = 6 \rightarrow z = \frac{6 - 3x - y}{2} = 3 - \frac{3}{2}x - \frac{1}{2}y$$

$$z_0 = f(2,1) = L(2,1) = 3 - 3 \cdot \frac{2}{2} - \frac{1}{2} \cdot 1 = -\frac{1}{2}$$

• Armar recta normal $\vec{r}_n(t)$

↪ Hallar normal al plano = $\vec{N} = (3, 1, 2)$ } $\vec{r}_n(t) = t(3, 1, 2) + (2, 1, -\frac{1}{2})$

↪ Hallar punto $A \rightarrow A = (2, 1, -\frac{1}{2})$

$$\vec{r}_n(t) \begin{cases} x = 3t + 2 \\ y = t + 1 \\ z = 2t - \frac{1}{2} \end{cases}$$

• Verificar $\vec{r}_n(t) \cap y = x^2$

$$(t+1) = (3t+2)^2$$

$$(t+1) = 3^2 t^2 + 12t + 4$$

$$0 = 9t^2 + 11t + 3$$

$$t_1, t_2 = \frac{-11 \pm \sqrt{11^2 - 4(9)(3)}}{18}$$

$$t_1 = \frac{-11 + \sqrt{13}}{18}$$

$$t_2 = \frac{-11 - \sqrt{13}}{18}$$

Hay dos intersecciones de la recta normal al sup $y = x^2$

* Ejercicio 24 $L: \mathbb{R}^n \rightarrow \mathbb{R} \xrightarrow{\sim} L(x) = \sum_{i=1}^n c_i x_i$, demostrar L diferenciable $\forall A \in \mathbb{R}^n$, $\nabla L(A) = (c_1, \dots, c_n)$

↪ y deriv. direc en A en $\vec{v} = \underline{L'(A, \vec{v}) = L(\vec{v})}$

• Demostrar diferenciable $\rightarrow$ Demostrar clase C^1

↪ Si L es función lineal entonces es clase C^∞ , diferenciable en su dom $\forall A \in D$ y $D \subset \mathbb{R}^n$

• Demostrar $\nabla L(A) = (c_1, c_2, \dots, c_n)$

↪ armar ∇L

$$\nabla L(A) = \left(\frac{\partial L}{\partial x_1}, \frac{\partial L}{\partial x_2}, \dots, \frac{\partial L}{\partial x_n} \right)$$

$$\frac{\partial L}{\partial x_1}(x) = c_1 \quad \frac{\partial L}{\partial x_2} = c_2 \quad \dots \quad \frac{\partial L}{\partial x_n} = c_n \quad , \text{ donde } \{c_1, \dots, c_n\} \in \mathbb{R}$$

$$\nabla L(A) = (c_1, c_2, \dots, c_n) \quad \forall A \in D \subset \mathbb{R}^n$$

• Demostrar $L'(A, \vec{v}) = L(\vec{v})$, $\vec{v} = (a_1, \dots, a_n) \in \mathbb{R}^n$

$$L'(A, \vec{v}) = \nabla L(A) \cdot \vec{v} = (c_1, c_2, \dots, c_n) (a_1, a_2, \dots, a_n) = \underbrace{c_1 a_1 + c_2 a_2 + \dots + c_n a_n}_{\sum_{i=1}^n c_i a_i = L(\vec{v})}$$

por lo que $L'(A, \vec{v}) = L(\vec{v})$

* Ejercicio 25: $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ diferenciable en todo $\mathbb{R}^2$

↳ plano tangente a sup en $A = (1, -\frac{1}{2}, f(1, -\frac{1}{2})) \Rightarrow 2x - 4y - 2z = 6$

↳ Hallar $f(1, -\frac{1}{2})$ y $\nabla f(1, -\frac{1}{2})$

• Hallar $f(1, -\frac{1}{2})$

Si el plano es tangente a la gráfica de la función, entonces, tienen el mismo punto $f(x, y) = L(x, y)$
en $(x, y) = (1, -\frac{1}{2})$

↳ Hallar $L(x, y)$

$$2x - 4y - 2z = 6 \rightarrow z = \frac{2x - 4y - 6}{2} = x - 2y - 3 \rightarrow z = L(x, y) = x - 2y - 3$$

↳ Hallar $f(1, -\frac{1}{2}) = L(1, -\frac{1}{2}) = 1 - 2(-\frac{1}{2}) - 3 = 1 + 1 - 3 = -1 \Rightarrow f(1, -\frac{1}{2}) = -1$

↳ Hallar $\nabla f(1, -\frac{1}{2}) = (\frac{\partial f}{\partial x}(1, -\frac{1}{2}), \frac{\partial f}{\partial y}(1, -\frac{1}{2})) = (1, -2)$
